

MIDLAND, TX, USA

WMO: 722324

Lat: 32.036N

Lon: 102.101W

Elev: 2803

StdP: 13.27

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03071

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.8	(c) 27.1	(d) 3.0	(e) 7.1	(f) 45.2	(g) 6.9	(h) 8.7	(i) 47.5	(j) 29.1	(k) 59.0	(l) 25.1	(m) 55.0	(n) 6.9	(o) 40	(p) 0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	102.6	66.7	100.2	67.1	98.1	67.1	73.1	87.1	72.3	86.5	71.4	85.9	10.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.2	123.4	77.0	68.9	117.9	76.7	67.9	113.9	76.8	38.7	86.8	37.8	86.3	37.2	85.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.2	(b) 20.6	(c) 18.6	(e) 16.6	(f) 106.5	(g) 4.5	(h) 3.4	(i) 13.4	(j) 109.0	(k) 10.8	(l) 110.9	(m) 8.3	(n) 112.8	(o) 5.0	(p) 115.3		
(a) 24.2	(b) 20.6	(c) 18.6	(e) 14.5	(f) 74.4	(g) 4.1	(h) 1.5	(i) 11.5	(j) 75.5	(k) 9.1	(l) 76.4	(m) 6.8	(n) 77.3	(o) 3.8	(p) 78.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.7	46.7	51.2	59.5	67.6	75.2	83.6	84.4	83.9	77.0	66.5	56.0	47.4
	DBStd	15.86	9.18	10.63	9.54	8.41	8.24	5.15	4.54	4.82	5.91	8.87	8.88	9.09
	HDD50	519	173	104	30	4	0	0	0	0	0	7	44	157
	HDD65	2191	568	394	218	69	18	0	0	0	3	91	285	545
	CDD50	6597	70	137	325	532	780	1007	1065	1051	810	517	225	78
	CDD65	2797	0	8	49	147	333	557	600	586	363	137	16	1
	CDH74	34624	29	165	676	1800	3969	7248	7825	7520	3810	1337	210	35
	CDH80	18269	1	32	207	762	2084	4183	4439	4245	1796	482	37	1
Wind	WSAvg	9.2	8.5	9.4	10.2	11.2	10.6	10.7	9.0	8.1	8.1	8.6	8.2	8.3
Precipitation	PrecAvg	14.0	0.6	0.9	0.8	0.8	1.7	1.8	1.8	1.8	1.7	1.3	0.8	0.5
	PrecMax	23.6	2.1	2.6	2.4	2.9	7.1	3.7	6.7	3.6	4.5	4.4	5.5	2.7
	PrecMin	6.4	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.2	0.0	0.0	0.0	0.0
	PrecStd	4.7	0.6	0.7	0.6	0.9	1.9	1.1	1.8	1.1	1.3	1.3	1.2	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.5	84.2	90.1	95.4	104.0	107.1	104.6	104.0	98.8	92.9	84.3	78.5
		MCWB	52.1	54.6	56.7	58.5	62.6	65.8	69.1	67.8	66.0	64.2	57.0	53.0
	2%	DB	72.7	79.0	85.8	91.6	99.0	103.2	101.1	101.1	95.6	89.6	79.4	72.8
		MCWB	50.1	53.0	55.6	58.1	60.8	65.5	68.4	68.2	66.5	63.2	57.1	51.6
	5%	DB	68.1	74.7	81.6	88.5	95.2	99.8	99.0	99.0	93.0	85.9	75.7	68.1
		MCWB	48.6	52.1	54.9	57.8	61.7	66.7	68.5	68.5	66.1	63.1	55.4	49.8
	10%	DB	63.4	69.9	77.3	84.5	91.1	96.9	96.5	96.4	89.9	81.8	72.1	63.2
		MCWB	47.5	50.0	54.3	57.6	61.9	67.2	68.3	68.6	66.6	62.0	54.4	48.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	60.2	64.3	68.0	72.4	74.1	74.5	73.9	73.2	72.1	64.4	59.7
		MCDB	65.1	70.6	74.3	80.1	83.0	88.8	86.2	88.3	85.0	83.9	70.3	66.1
	2%	WB	53.9	57.6	62.2	65.9	70.4	72.8	72.7	72.7	71.7	69.5	61.8	55.8
		MCDB	65.2	69.3	72.4	76.4	81.9	88.0	86.0	88.1	83.7	78.4	69.9	63.3
	5%	WB	50.7	54.7	60.4	64.1	69.0	71.8	71.9	71.8	70.7	68.0	59.8	52.8
		MCDB	65.2	69.1	71.0	74.9	80.7	86.3	86.2	87.5	82.5	77.9	68.6	64.7
	10%	WB	48.0	52.0	58.4	62.2	67.7	70.8	71.1	71.0	69.7	66.1	57.5	49.8
		MCDB	62.1	67.0	70.3	74.4	80.3	85.6	86.3	86.7	81.0	76.4	68.5	63.0
Mean Daily Temperature Range	5% DB	MDBR	23.7	25.0	25.9	26.4	24.2	23.7	21.5	21.6	21.1	23.0	24.2	22.7
		MCDBR	30.9	32.0	30.7	30.8	28.9	27.4	24.4	24.2	24.9	26.5	28.2	30.3
		MCWBR	14.3	14.2	12.1	11.3	10.2	7.1	6.4	5.8	7.0	9.7	12.1	14.6
	5% WB	MCDBR	27.2	27.9	24.5	23.9	23.7	23.2	20.9	21.3	19.3	20.6	22.1	24.9
		MCWBR	14.3	14.5	12.4	11.6	10.3	7.2	6.0	6.0	6.2	9.3	12.4	14.2
Clear-Sky Solar Irradiance	taub	0.266	0.277	0.297	0.333	0.350	0.379	0.396	0.373	0.354	0.316	0.288	0.270	
	taud	2.575	2.545	2.502	2.396	2.415	2.378	2.377	2.413	2.434	2.524	2.536	2.561	
	Ebn at Noon	310	314	313	303	297	286	281	287	290	296	298	301	
	Edh at Noon	25	29	32	38	37	39	39	37	35	29	26	24	
All-Sky Solar Radiation	RadAvg	1097	1356	1756	2150	2279	2409	2285	2119	1764	1444	1150	978	
	RadStd	107	143	127	134	164	114	127	94	122	162	121	89	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	+135

Nomenclature: See separate page